

Mohit Varma Sagi

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Skills & Certifications

Languages: Python (Pandas, NumPy, Scikit-learn, Pdfplumber), SQL, T-SQL, PostgreSQL

ML & Modeling: XGBoost, PyTorch, SHAP, LSTM, CNN, Predictive Modeling, Regression Analysis, Feature Engineering, Statistical Modeling

Generative AI & LLMs: LangChain, LangGraph, RAG Pipelines, Agentic AI, ChromaDB, FAISS, Vector Databases, Hugging Face

Data Engineering: ETL/ELT, Azure Data Factory, Azure Logic Apps, Power Automate, Microsoft Fabric, Databricks, Spark, MLflow, AWS (Lambda, EC2, S3), RPA

Analytics & BI: Power BI, DAX, Tableau, Stakeholder Reporting, Automated Reporting Pipelines, Data Storytelling

Certifications: ML with Python (MIT 6.86x) · Intro to CS (MIT 6.00.1x) · Data Science CDS (Henry Harvin)

Experience

Data Analyst (AI & Automation, RCM) | EyeSouth Partners

Aug 2025 – Present

Atlanta, GA

- Reduced prior authorization research from 3 hours to 5 minutes by building a production RAG tool with LangChain and ChromaDB across 1,000+ payer policy documents.
- Identified \$2.4M+ in A/R recovery opportunities by engineering ETL/ELT pipelines on Azure Data Factory and Microsoft Fabric across 300+ providers, replacing manual aging reports with daily automated refreshes.
- Designed and maintained RCM analytics infrastructure including SQL stored procedures, three-layer data pipelines, and Power BI dashboards across 44 locations covering denials, AR aging, and pending encounters.
- Partnered with Practice Admins, Regional Directors, and Client Success Managers to gather requirements and deliver the Doctor Package – automated per-provider PDF exports via Logic App and Office 365.
- Built a Python EOB extraction tool using pdfplumber and pandas to process payer remittance PDFs across multiple formats, extracting CPT codes, billed and allowed amounts, and reversals with AI-assisted validation.

Data Analyst, RCM (Intern) | EyeSouth Partners

May 2025 – Aug 2025

Atlanta, GA

- Surfaced \$200K+ in recoverable payer underpayments by training an XGBoost model on 2 years of claims data with SHAP-based feature selection, enabling targeted follow-up workflows.
- Built a Power BI KPI dashboard tracking pending encounters and unbilled claims across affiliates, cutting stakeholder reporting turnaround by 3 days.
- Automated extraction of 10,000+ billing codes in Python, reducing manual errors by 90% and enabling self-service reporting across 3 business units.

Research Scientist, Applied Data Science | Southeast Regional Cooperatives

Jan 2025 – May 2025

Atlanta, GA

- Trained a hybrid LSTM and CNN forecasting model on 20,000+ records in Azure Databricks with MLflow tracking, surfacing 3 yield-driving patterns that informed seasonal planning.
- Built a GPT-4 RAG pipeline across 15 research datasets, cutting literature lookup time by 60% for research teams.

Research Analyst, ESG Data Analytics | Truist Financial

Aug 2024 – Dec 2024

Atlanta, GA

- Eliminated 40 hours/month of manual reporting by building an AWS Lambda ETL pipeline and Tableau dashboards for ESG strategy reporting across 6 financial institutions.
- Analyzed 200+ ESG and climate risk metrics in Python using statistical modeling to expose regulatory disclosure gaps, with findings presented to board-level leadership.

Operations Analyst Intern | Hindustan Petroleum Corporation Limited

May 2023 – Jul 2023

India

- Deployed IoT anomaly detection on AWS EC2 using Z-score and IQR thresholds, flagging equipment failures across 10 machines before downtime occurred.
- Applied STL time-series decomposition on 6 months of sensor data, identifying 3 recurring failure patterns that optimized preventive maintenance schedules.

Projects

Multi-Agent Research Copilot

- Architected a 3-agent LangGraph system covering query decomposition, hybrid BM25 and vector retrieval, and answer synthesis across 200+ PDFs with cited RAG responses.

Predictive Analytics & Risk Modeling Platform

- Built a stacked ensemble model on 50,000+ records achieving 89% accuracy with SHAP explainability and MLflow tracking; delivered a Power BI risk-tier dashboard for stakeholder prioritization.

Education

M.S. in Data Science and Analytics | Georgia State University, J. Mack Robinson College of Business

Dec 2025

B.Tech. in Production Engineering and Management | National Institute of Technology Calicut

May 2024